Analysis and Design of Algorithms

NUS National University of Singapore

CS3230

Week 8

Dynamic Programming

Diptarka Chakraborty

Ken Sung

Recall: Divide and Conquer Paradigm

- **Divide** the input instance into 2 or more parts (of equal sizes).
- Solve the same problem for each smaller instance **recursively**.
- **Combine** their solutions to get the solution of the original instance.

It is usually quite straight forward to see if divide and conquer can be applied to solve a problem.

Fibonacci Number $F(n)$

- $F(0) = 0$
- $F(1) = 1$
- $F(n) = F(n-1) + F(n-2)$ for $n>1$

Problem: Given n compute $F(n)$

- Recursive algorithm
- Iterative algorithm

Analyze recursive algorithm for F(n)

```
RFIB(n)
{
        if n=0
                return 0;
        else if n=1
                return 1;
        else
                return((RFIB(n-1) + RFIB(n-2));
}
```

Let T(n) be the number of instructions executed by **RFIB**(n)

$T(0) = 1$

$T(1) = 2$

$T(n) = 4 + T(n-1) + T(n-2)$ for $n>1$

Simple Exercise:

Show $F(n) > 2^{(n-2)/2}$

(may be use induction)

Observation: $T(n) > F(n)$

Analyze iterative (array-filling) algorithm for F(n)

```
IFIB(n) {
        F[0] ← 0;
        F[1] ← 1;
        For(i=2 to n) do
            F[i] ← F[i-1] + F[i-2];
        return F[n];
}
```

2 instructions

$n - 1$ iterations

1 instruction

1 instruction

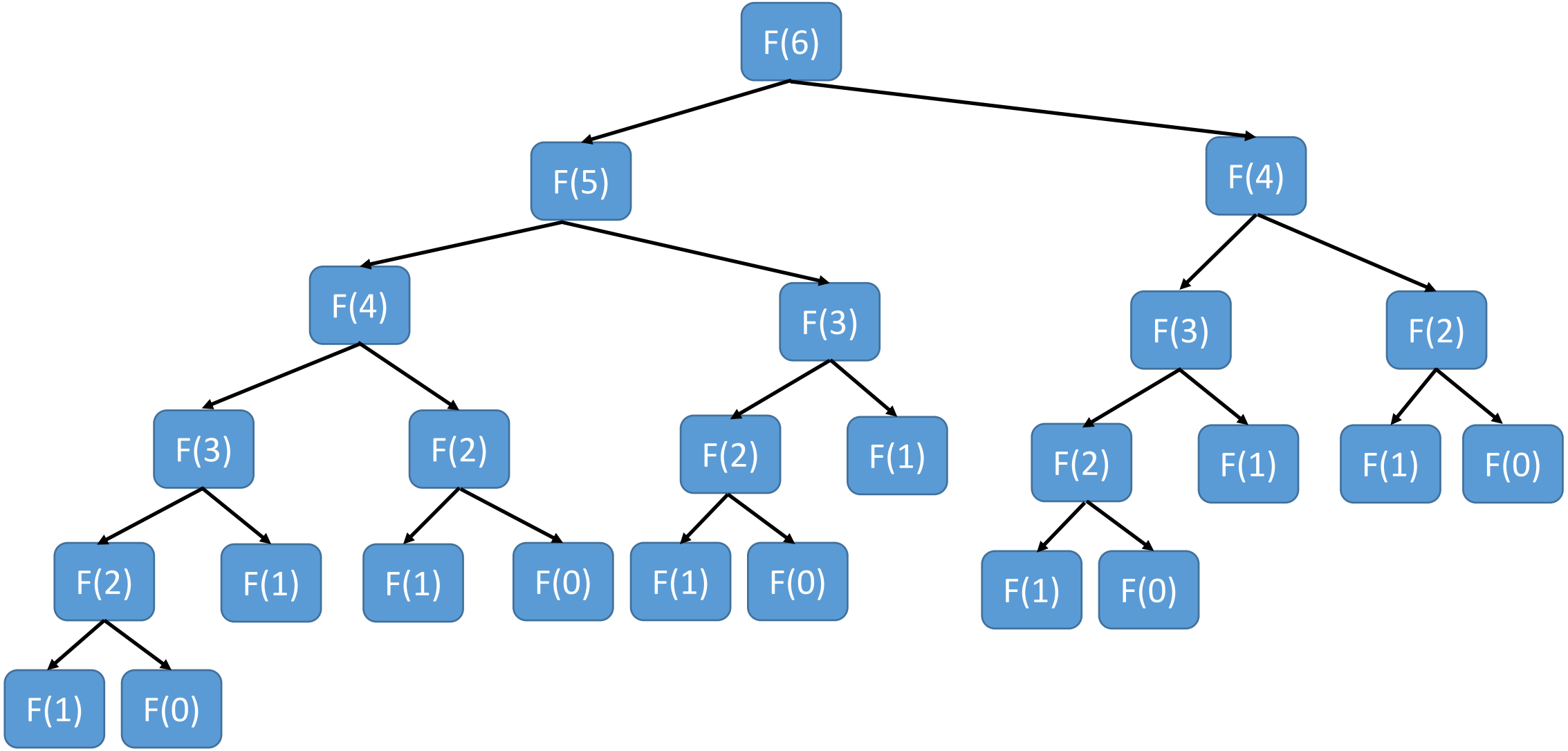
No. of instructions = $2 + (n - 1) \times 1 + 1 = n + 2$

Compare two algorithms for F(n)

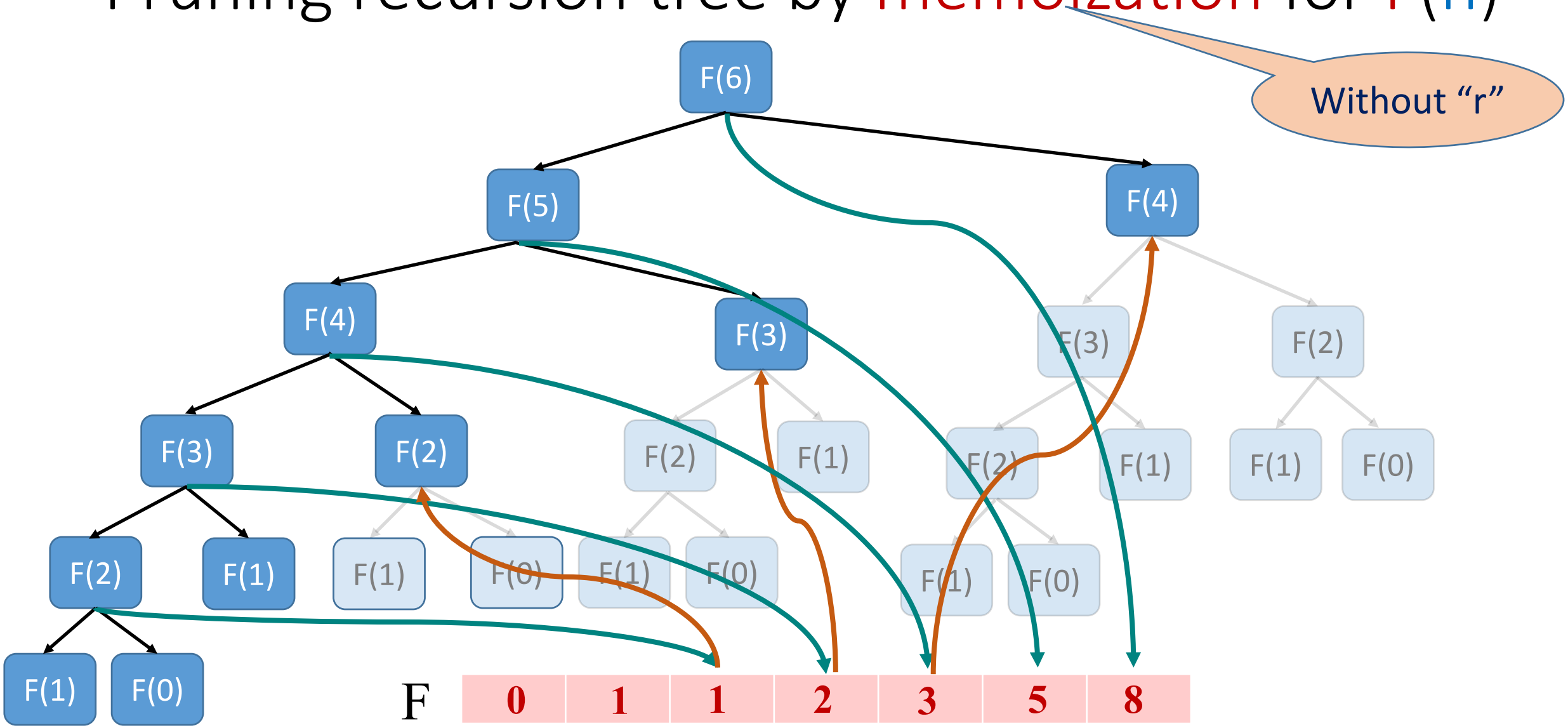
- No. of instructions by recursive algorithm **RFIB**(n) is $> 2^{(n-2)/2}$ (exponential in n)
- No. of instructions by iterative algorithm **IFIB**(n) is = $n + 2$(linear in n)

Can you see why IFIB() is much faster than RFIB() ?

Recursion tree for F(n)



Pruning recursion tree by memoization for F(n)



Longest Common Subsequence

Applications in Computational Biology, Text Processing and many more

What is a subsequence?

Sequence $A : a_1, a_2, \ldots, a_n$

Can be stored in an array $A[1..n]$, $A[j]$: a_j $\qquad$ $A[1..k]$: $a_1, a_2, \ldots, a_k$

Definition: C is said to be a **subsequence** of A if we can obtain C by removing 0 or more elements from A.

Example: $A : a, \cancel{b}, a, \cancel{d}, e, b, \cancel{c}, a$

$C : a, a, e, b, a$

A more formal definition:

C is a **subsequence** of A if there exists k integers: $1 \le i_1 < \cdots < i_k \le n$ s.t.

for all $1 \le j \le k$ $\qquad C[j] = A[i_j]$

Longest Common Subsequence - Problem Definition

Given : Two sequences $A[1..n]$ and $B[1..m]$,

Aim : To compute a (not "the") <u>longest</u> sequence C s.t.

C is subsequence of A as well as B

Example: A : $a\ a\ s\ b\ d\ e\ s\ c\ b\ d$

B : $a\ c\ b\ s\ d\ c\ d\ e\ b$

Answer: $a\ s\ d\ e\ b$

Question : How to compute a LCS of A and B efficiently.

Finding LCS: Trivial Brute-Force Solution

Given : two sequences $A[1..n]$ and $B[1..m]$

$$A : a_1, a_2, \ldots, a_n$$

$$B : b_1, b_2, \ldots, b_m$$

- Check all the possible subsequences of A to see if it is also a subsequence of B, and then output the longest one.

Analysis:

- Checking whether a particular subsequence of A is a subsequence of B takes $O(m)$ time.
- How many possible subsequences of A are there? 2^n

  (Each bit-vector of length n determines a distinct subsequence)
- So total time = $O(m2^n)$

Can we do better?

Finding LCS: Recursive Formulation

Given : two sequences $A[1..n]$ and $B[1..m]$

$$A : a_1, a_2, \ldots, a_n$$

$$B : b_1, b_2, \ldots, b_m$$

Notation for recursive formulation:

LCS(i, j): Longest common subsequence of $A[1..i]$ and $B[1..j]$

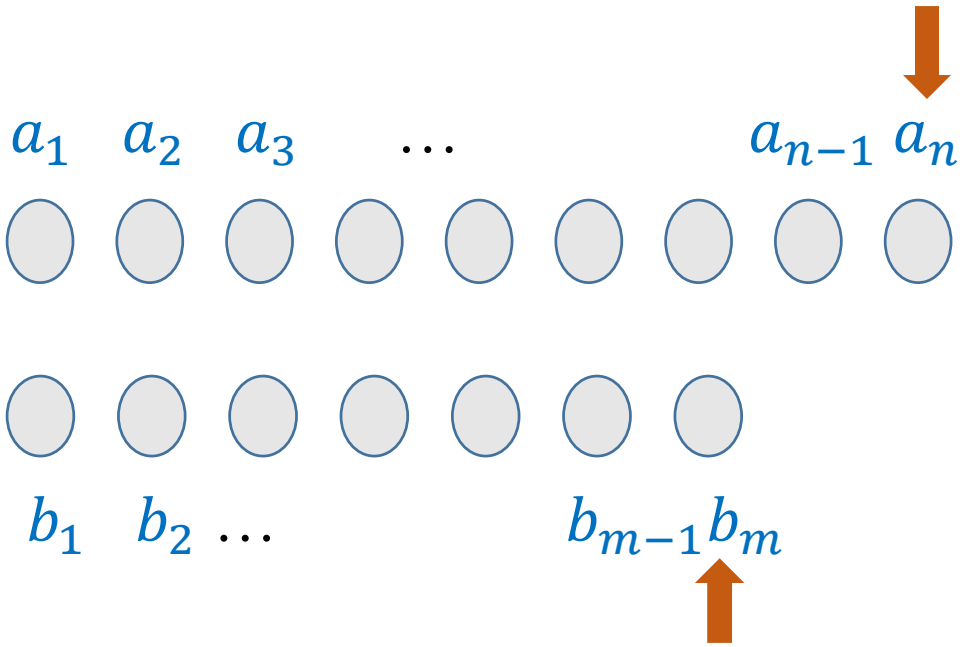
Aim : To express **LCS**(i, j) recursively.

Base Case:

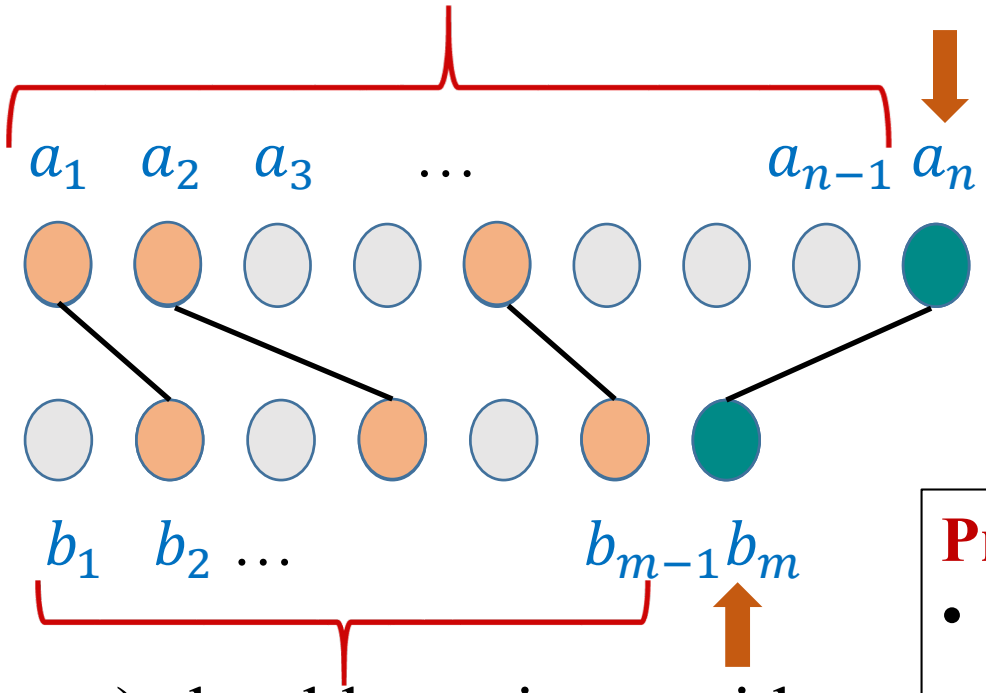
$$\left.\begin{array}{ll} \textbf{LCS}(i, 0) = \emptyset & \text{for all } i \\ \textbf{LCS}(0, j) = \emptyset & \text{for all } j \end{array}\right\}$$

Since one of the sequences is **empty**

Recursive Formulation of LCS(n,m)



How does LCS(n,m) look like when $a_n = b_m$?



Intuition: **LCS**(n,m) should terminate with a_n

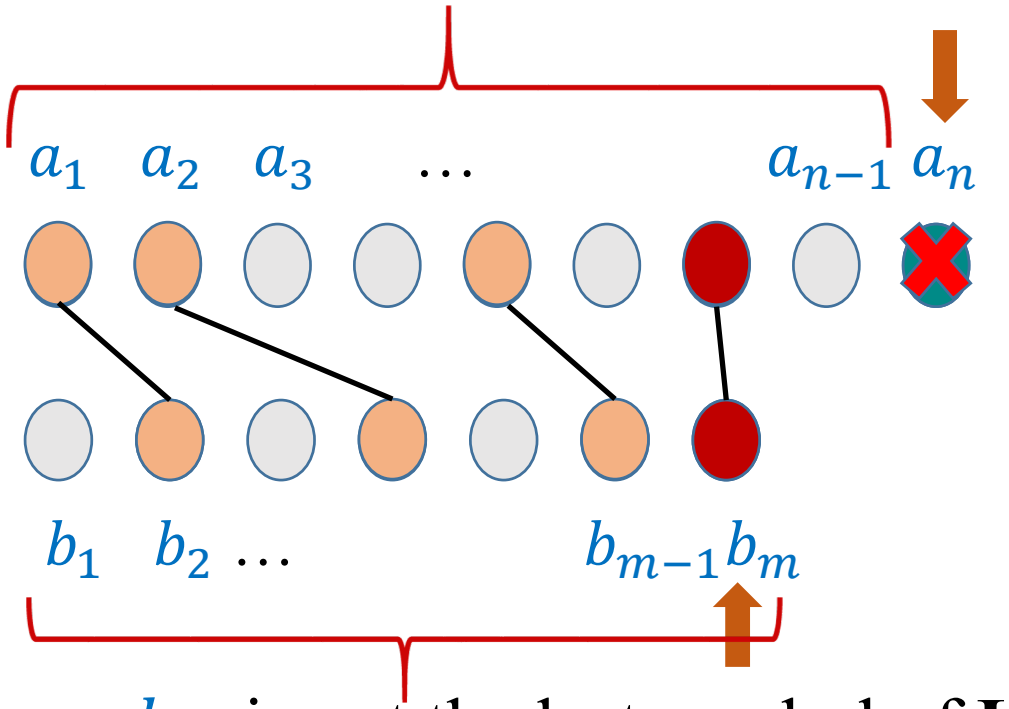
Lemma: If $a_n = b_m$ then

LCS(n,m) = **LCS**($n-1$,$m-1$) :: a_n

Proof Idea:

- LCS(n,m) must terminate with the symbol same as a_n; otherwise we could <u>extend</u> the solution by concatenating a_n
- Observe, it is fine to <u>*match*</u> a_n with b_m

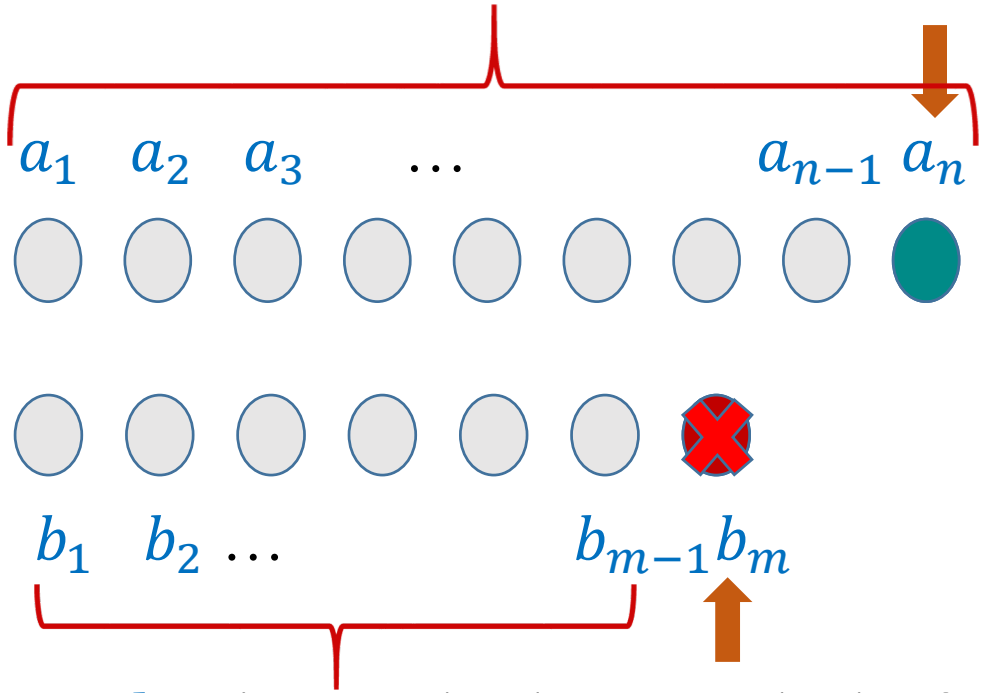
How does LCS(n,m) look like when $a_n \neq b_m$?



Intuition: Either a_n or b_m <u>is not</u> the last symbol of **LCS**(n,m)

Observation: If a_n <u>is not</u> the last symbol of **LCS**(n,m) **LCS**(n,m) = **LCS**($n-1$,m)

How does LCS(n,m) look like when $a_n \neq b_m$?

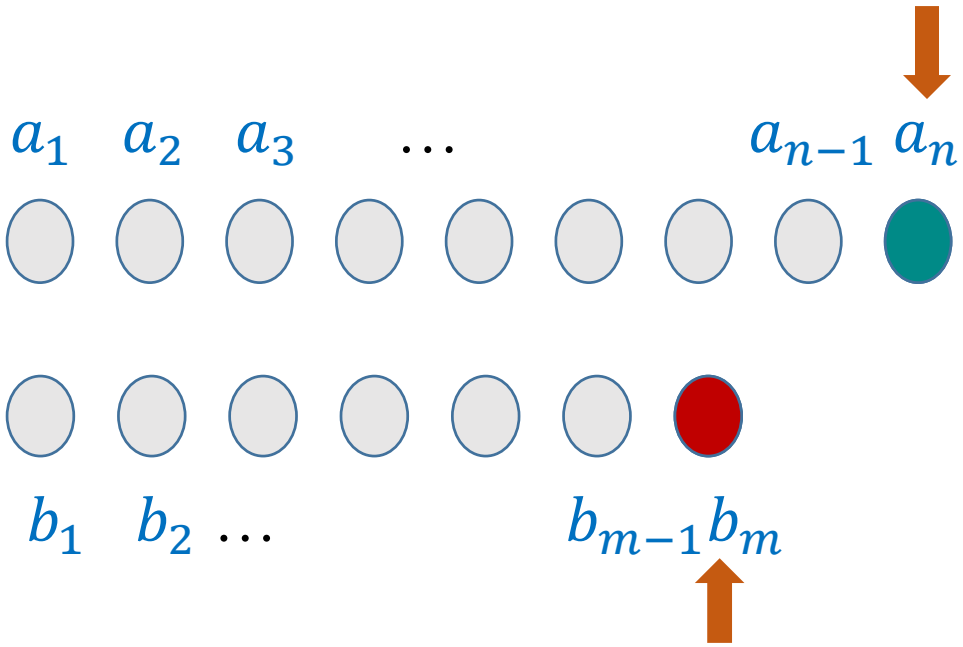


Intuition: Either a_n or b_m <u>is not</u> the last symbol of **LCS**(n,m)

Observation: If a_n <u>is not</u> the last symbol of **LCS**(n,m) **LCS**(n,m) = **LCS**($n-1$,m)

Observation: If b_m <u>is not</u> the last symbol of **LCS**(n,m) **LCS**(n,m) = **LCS**(n,$m-1$)

How does LCS(n,m) look like when $a_n \neq b_m$?



Intuition: Either a_n or b_m <u>is not</u> the last symbol of **LCS**(n,m)

Lemma: If $a_n \neq b_m$ then
LCS(n,m) is either **LCS**($n-1$,m) or **LCS**(n,$m-1$)

Finding LCS: Recursive Formulation

Optimal substructure

An optimal solution to a problem (instance) contains optimal solutions to subproblems

Base Case:

LCS(i, 0) = ∅ for all i

LCS(0, j) = ∅ for all j

General Case:

If $a_n = b_m$ then **LCS**(n,m) = **LCS**($n-1$,$m-1$) :: a_n

If $a_n \neq b_m$ then **LCS**(n,m) = <u>*bigger*</u> of **LCS**(n,$m-1$) or **LCS**($n-1$,m)

Simplified Problem: Find the length of LCS

Let $L(n,m)$: Length of LCS of $A[1..n]$ and $B[1..m]$

$L(n,m) = 0$ if n or m is 0

Question 1

Can you represent $L(n,m)$ in terms of subproblems?

Finding LCS: Recursive Formulation

Optimal substructure

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If $a_n = b_m$ then **LCS**(n,m) = **LCS**($n-1$,$m-1$) :: a_n

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Simplified Problem: Find the length of LCS

Let $L(n,m)$: Length of LCS of $A[1..n]$ and $B[1..m]$

$L(n,m) = 0$ if n or m is 0

If $a_n = b_m$ then $L(n,m) = \mathrm{L}(n-1,m-1) + 1$

If $a_n \neq b_m$ then $L(n,m) = \mathbf{Max}(\mathrm{L}(n,m-1), \mathrm{L}(n-1,m))$

Recursive algorithm for $L(n,m)$

```
L(n,m)
{   If (n = 0 or m = 0)
        return 0;
    Else
    {   If a_n = b_m then
            return (L(n − 1,m − 1) + 1);
        Else
        {   l_1 ← L(n − 1,m) ;
            l_2 ← L(n,m − 1) ;
            return Max(l_1, l_2);
        }
    }
}
```

$\boldsymbol{T}(n,m)$: Worst case running time of $L(n,m)$

$\boldsymbol{T}(n,m) = \boldsymbol{T}(n-1,m) + \boldsymbol{T}(n,m-1)$

A simple exercise from discrete math:

$\boldsymbol{T}(n,m) \geq \binom{n+m}{n} > 2^n$ (assuming $\boldsymbol{m} \approx \boldsymbol{n}$)

Exponential !!

But why ?

Let us explore

Recursive algorithm for $L(n,m)$

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L(n,m)
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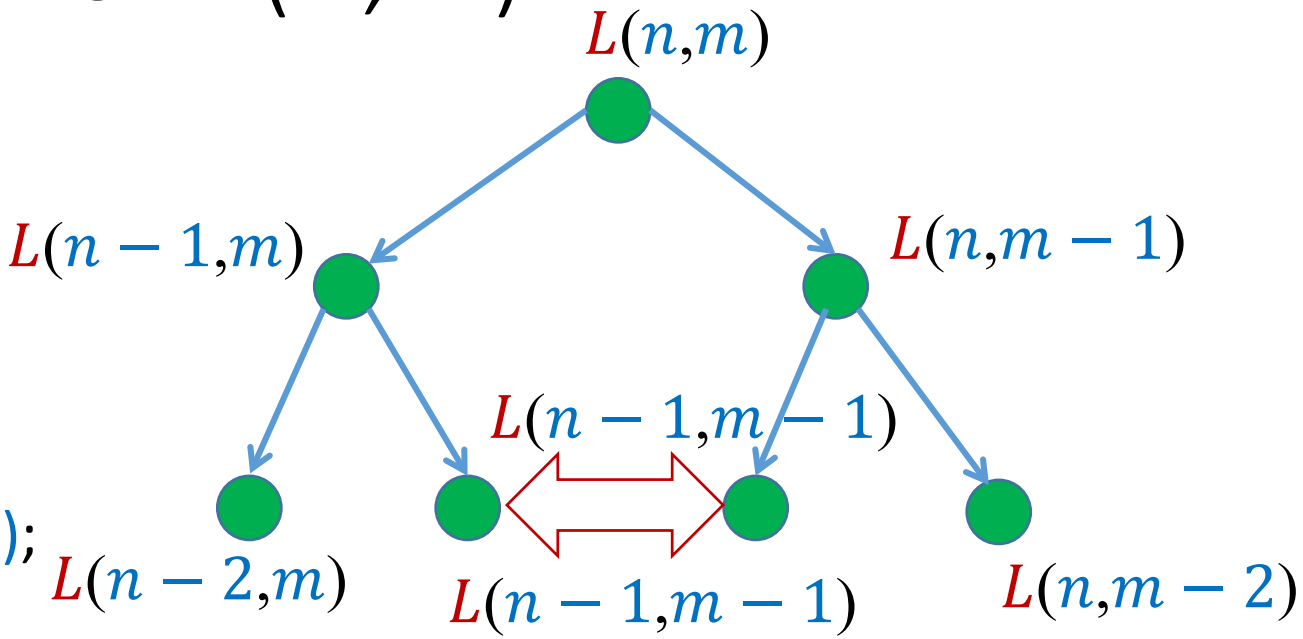
        return 0;
    Else
    {   If a_n = b_m then
            return (L(n − 1,m − 1) + 1);
        Else
        {   l_1 ← L(n − 1,m) ;
            l_2 ← L(n,m − 1) ;
            return Max(l_1, l_2);
        }
    }
}
```



- Solving same sub-problem multiple times !!
- But how many **distinct** sub-problems are there ?
- Only $(n+1) \times (m+1)$

Overlapping subproblems

A recursive solution contains a "small" number of distinct subproblems repeated many times

Recursive algorithm for $L(n,m)$

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    {   If a_n = b_m then
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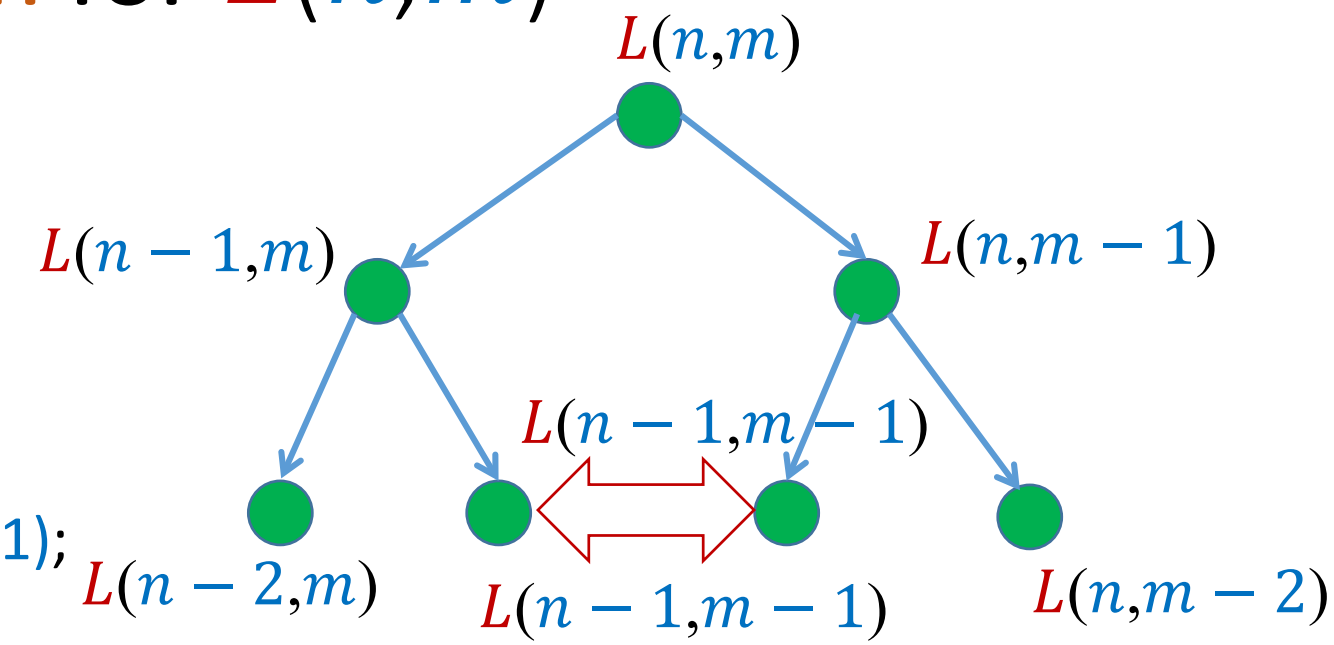

- Solving same sub-problem multiple times !!
- But how many sub-problems are there ?
- Only $(n+1) \times (m+1)$
- Can we compute them efficiently ?
- Get inspiration from algo for Fibonacci number !

Recursive algorithm for $L(n,m)$

```
L(n,m)
{   If (n = 0 or m = 0)
        return 0;
    Else
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        {   l_1 ← L(n − 1,m) ;
            l_2 ← L(n,m − 1) ;
            return Max(l_1, l_2);
        }
    }
}
```

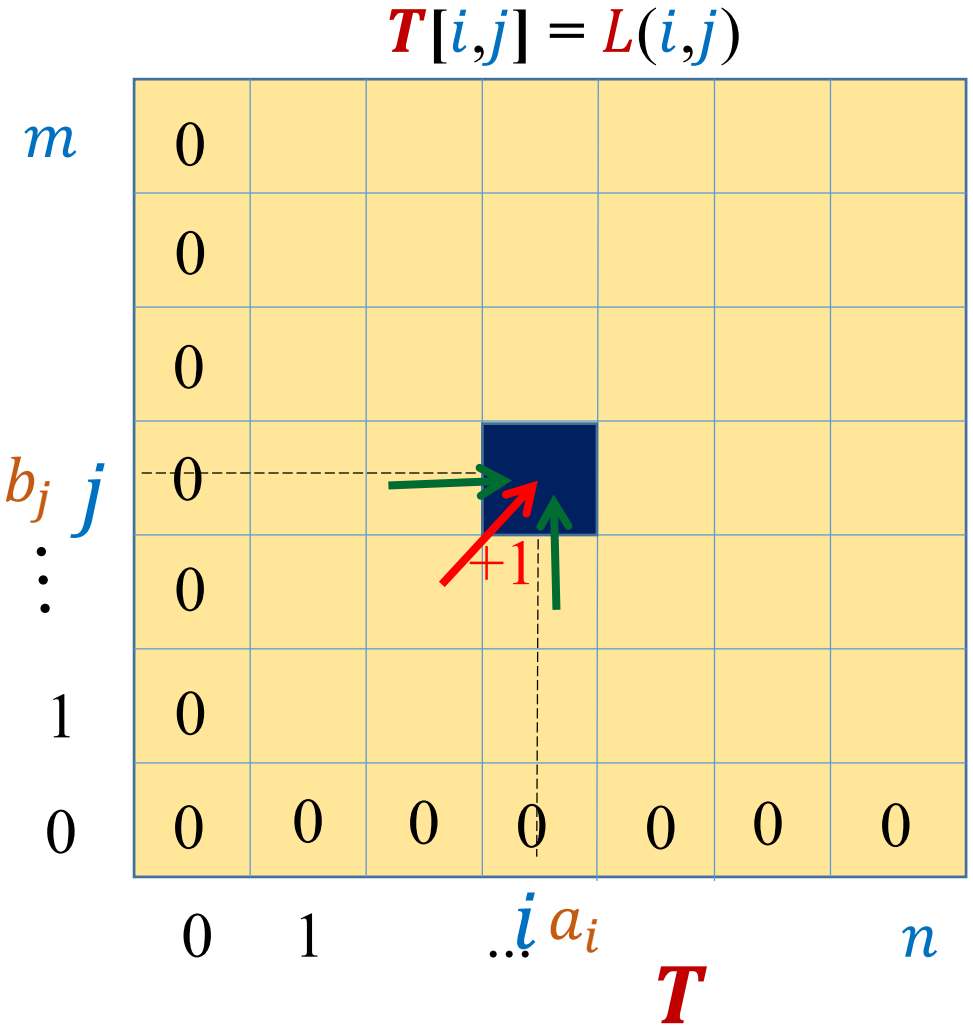
$T[i,j] = L(i,j)$

m	0						
	0						
	0						
	0						
$\vdots$	0						
1	0						
0	0	0	0	0	0	0	0
	0	1	...				n

T

Dynamic Programming algorithm for $L(n,m)$

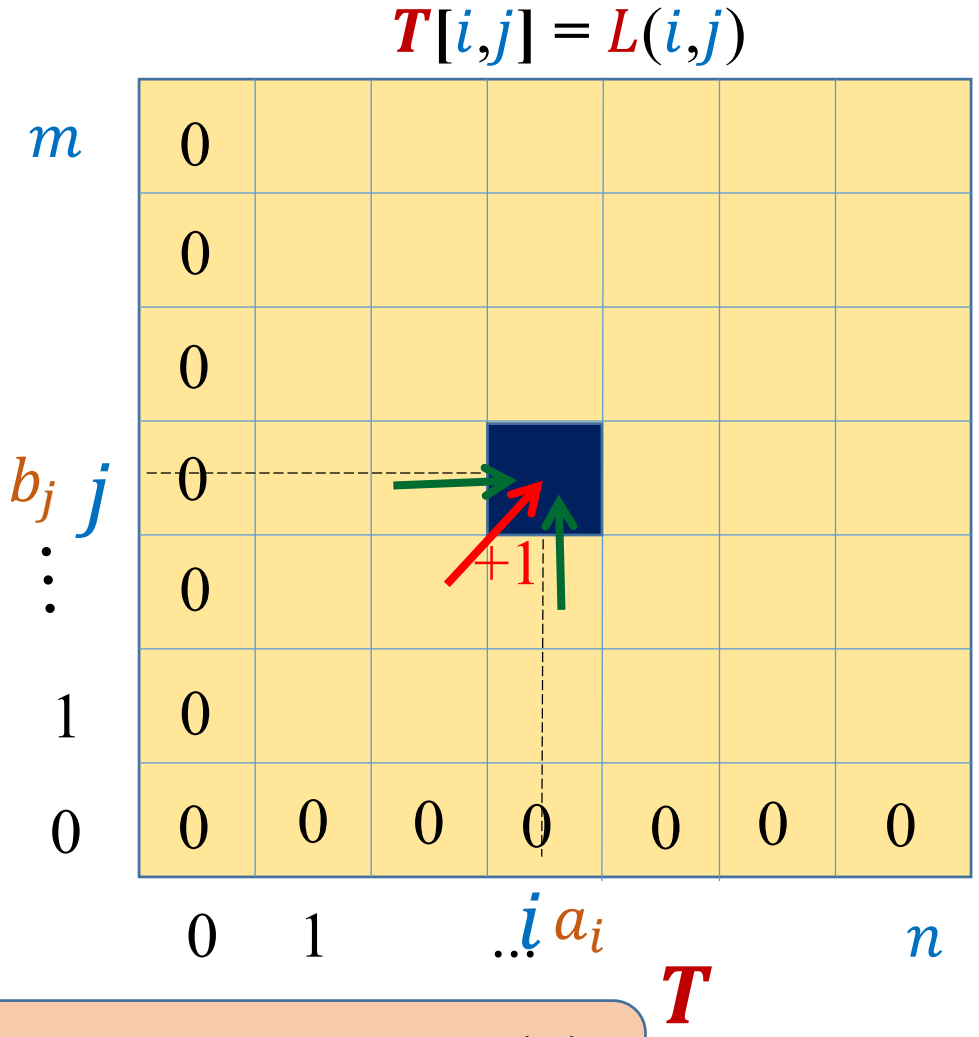
```
L(n,m)
{   For (i = 0 to n)   T[i,0] ← 0;
    For (j = 0 to m)   T[0,j] ← 0;
    For (j = 1 to m){
        For (i = 1 to n){
            If a_i = b_j then
                T[i,j] ← T[i − 1,j − 1] +1;
            Else {
                l_1 ←T[i − 1,j] ;
                l_2 ←T[i,j − 1] ;
                T[i,j] ← Max(l_1, l_2);
            } } }
}
```



Dynamic Programming algorithm for $L(n,m)$

$L(n,m)$

```
{   For (i = 0 to n)   T[i,0] ← 0;
    For (j = 0 to m)   T[0,j] ← 0;
    For (j = 1 to m){
        For (i = 1 to n){
            If a_i = b_j then
                T[i,j] ← T[i − 1,j − 1] +1;
            Else {
                l_1 ←T[i − 1,j] ;
                l_2 ←T[i,j − 1] ;
                T[i,j] ← Max(l_1, l_2);
            } } }
}
```



Time per table entry = $O(1)$
Total time = $O(nm)$

Dynamic Programming algorithm for $L(n,m)$

```
L(n,m)
{   For (i = 0 to n)   T[i,0] ← 0;
    For (j = 0 to m)   T[0,j] ← 0;
    For (j = 1 to m){
        For (i = 1 to n){
            If a_i = b_j then
                T[i,j] ← T[i − 1,j − 1] +1;
            Else {
                l_1 ←T[i − 1,j] ;
                l_2 ←T[i,j − 1] ;
                T[i,j] ← Max(l_1, l_2);
            } } }
}
```

$T[i,j] = L(i,j)$

a	0	1	2	2	3	3	4
d	0	1	2	2	3	3	3
c	0	1	1	2	2	2	3
a	0	1	1	1	2	2	3
d	0	0	1	1	2	2	2
d	0	0	1	1	1	1	1
	0	0	0	0	0	0	0
		a	d	c	d	s	a

T

Dynamic Programming algorithm for $L(n,m)$

```
L(n,m)
{   For (i = 0 to n)   T[i,0] ← 0;
    For (j = 0 to m)   T[0,j] ← 0;
    For (j = 1 to m){
        For (i = 1 to n){
            If a_i = b_j then
                T[i,j] ← T[i − 1,j − 1] +1;
            Else {
                l_1 ←T[i − 1,j] ;
                l_2 ←T[i,j − 1] ;
                T[i,j] ← Max(l_1, l_2);
            } } }
}
```

Note, you need to store the table T. So space requirement is $O(mn)$.

Exercise:

- How can you reduce the space requirement to $O(\min\{m,n\})$?
- Can you modify the algorithm so that you can output a LCS ?

Dynamic Programming algorithm paradigm

- Expressing the solution recursively
- Overall there are only polynomial number of subproblems
- But there is a huge overlap among the subproblems. So the recursive algorithm takes exponential time (solving same subproblem multiple times)
- So we compute the recursive solution iteratively in a bottom-up fashion (like in case of Fibonacci numbers). This avoids wastage of computation and leads to an efficient implementation

Question 2

Who is the Master of Algorithms pictured below?

- John Hopcroft
- Andrew Chi-Chih Yao
- Richard Bellman
- Ron Rivest

Question 2: Solution

Richard E. Bellman

Bellman started the systematic study of dynamic programming, providing the area with a solid mathematical basis. His contributions are foundational to control theory and economics.

Question 3

A palindrome is a string that reads the same forwards and backwards. Examples: *racecar*, *civic*, *a*.

Use the LCS algorithm to give an $O(n^2)$-time algorithm to find the longest palindrome subsequence of an input string.

Example: if input is *character*, output should be *carac*.

Question 3: Solution

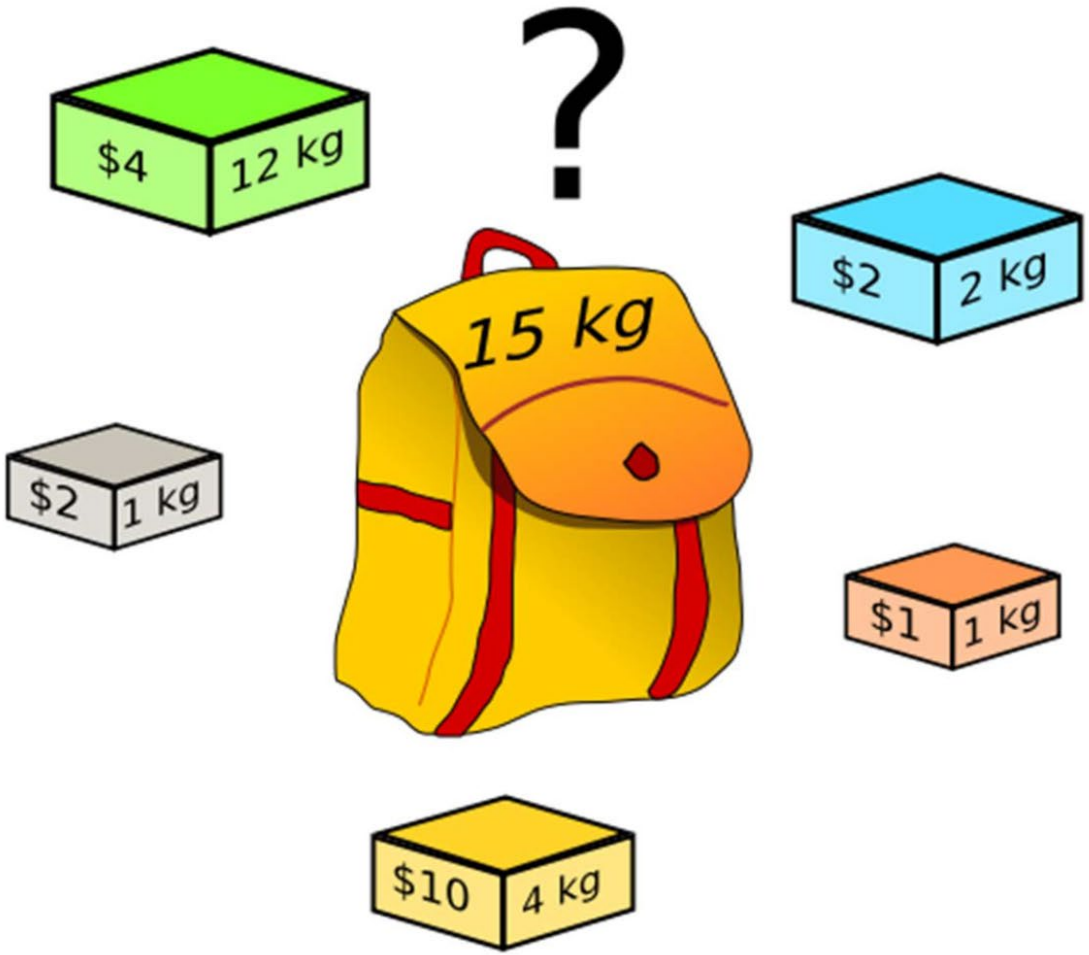
Run the LCS algorithm on the <u>input string and its reverse</u>!

More direct DP approach can also be formulated.

Exercise: Solve the problem this way.

Knapsack Problem

Knapsack Problem



What is the maximum value you can get?

Formal Definition

KNAPSACK

Input:

$(w_1, v_1), (w_2, v_2), \ldots, (w_n, v_n)$, and W

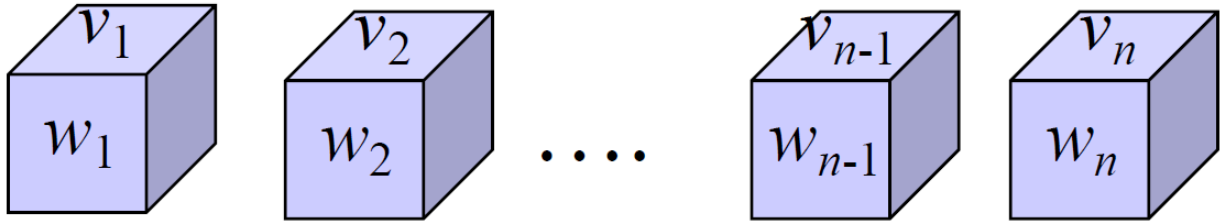
Output: A subset $S \subseteq \{1, 2, \ldots, n\}$ that maximizes

$\Sigma_{i \in S} v_i$ such that $\Sigma_{i \in S} w_i \leq W$

2^n many subsets, so naïve algorithm is too costly!

Dynamic Programming

Problem: $(w_1, v_1), \ldots, (w_n, v_n), W$

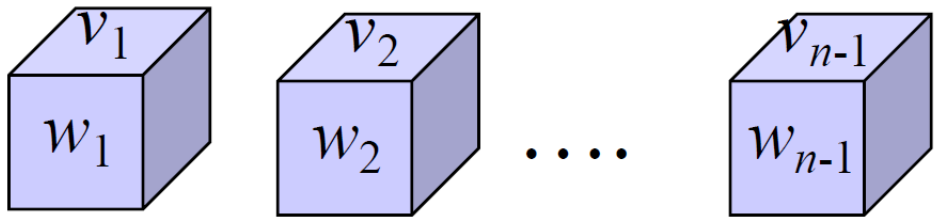


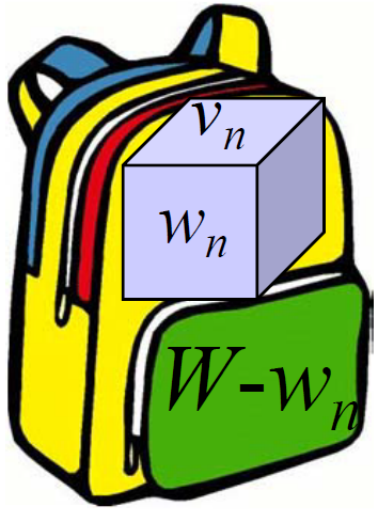
Is there optimal substructure?

Dynamic Programming

Case 1: Item n (the last one) is taken

Have optimal solution to subproblem defined by $(w_1, v_1), \ldots, (w_{n-1}, v_{n-1}), W\text{-}w_n$

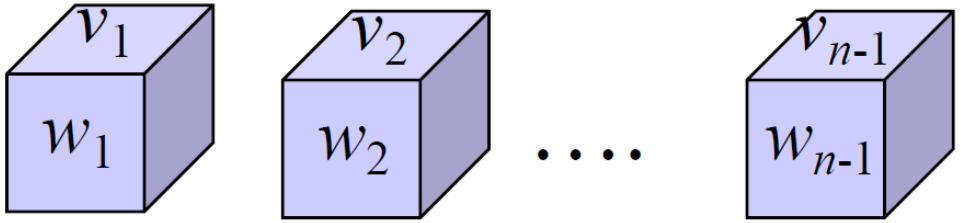




Dynamic Programming

Case 2: Item n (the last one) is **not** taken

Have optimal solution to subproblem defined by $(w_1, v_1), \ldots, (w_{n-1}, v_{n-1}), W$



Otherwise, by *cut and paste* argument, we can get a better solution

Recursive Solution

Let $m[i, j]$ be the maximum value that can be obtained using:

- a subset of items in $\{1,2, \dots, i\}$
- with total weight no more than j

$$m[i,j] = \begin{cases} 0, & \text{if } i = 0 \text{ or } j = 0 \\ \max\{m[i-1, j-w_i] + v_i, m[i-1,j]\}, & \text{if } w_i \leq j \\ m[i-1,j], & \text{otherwise} \end{cases}$$

Pseudocode

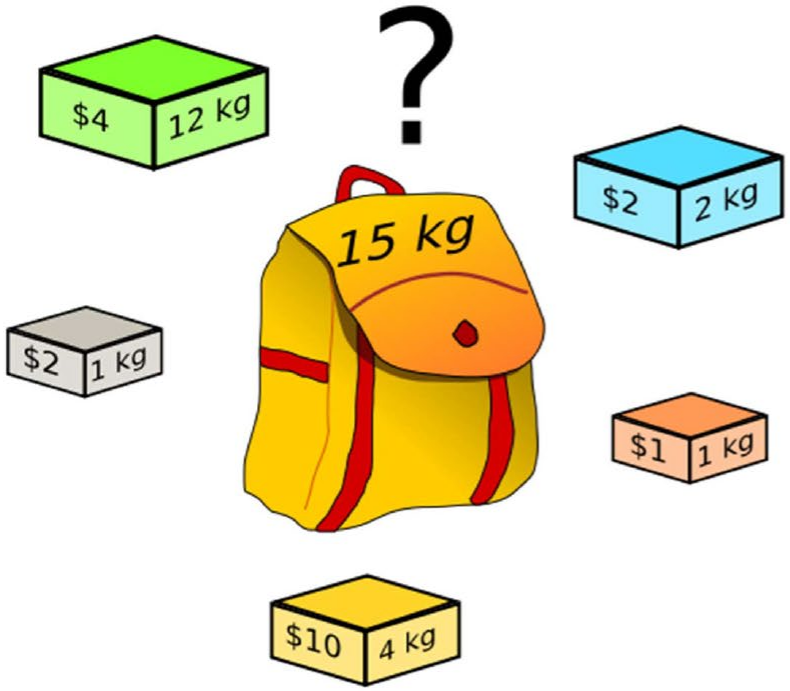
```
KNAPSACK(v, w, W):
    for j = 0, ..., W:
        m[0, j] ← 0
    for i = 1, ..., n:
        m[i, 0] ← 0
    ⟨Recursive cases⟩

    return m[n, W]
```

Pseudocode

```
⟨Recursive cases⟩

for i = 1, ..., n:
    for j = 0, ..., W:
        if j ≥ w[i]:
            m[i, j] ← max(m[i − 1, j − w[i]] + v[i], m[i − 1, j])
        else:
            m[i, j] ← m[i − 1, j]
```



$v[1..5] = \{4, 2, 10, 1, 2\}$

$w[1..5] = \{12, 1, 4, 1, 2\}$

$W = 15$

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0															
2	0															
3	0															
4	0															
5	0															

$$m[i,j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ \max\{m[i-1, j-w_i] + v_i, m[i-1, j]\} & \text{if } w_i \le j \\ m[i-1, j] & \text{otherwise} \end{cases}$$

$v[1..5] = \{4, 2, 10, 1, 2\}$

$w[1..5] = \{12, 1, 4, 1, 2\}$

$W = 15$

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	**4**			
2	0															
3	0															
4	0															
5	0															

$$m[i,j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ \max\{m[i-1, j-w_i] + v_i, m[i-1, j]\} & \text{if } w_i \le j \\ m[i-1, j] & \text{otherwise} \end{cases}$$

$v[1..5] = \{4, 2, 10, 1, 2\}$

$w[1..5] = \{12, 1, 4, 1, 2\}$

$W = 15$

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4	4
2	0	2	2	2	2	2	2	2	2	2	2	2	4	6	6	6
3	0	2	2	2	10	**12**										
4	0															
5	0															

$$m[i,j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ \max\{m[i-1, j - w_i] + v_i, m[i-1, j]\} & \text{if } w_i \le j \\ m[i-1, j] & \text{otherwise} \end{cases}$$

$$v[1..5] = \{4, 2, 10, 1, 2\}$$

$$w[1..5] = \{12, 1, 4, 1, 2\}$$

$$W = 15$$

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4	4
2	0	2	2	2	2	2	2	2	2	2	2	2	4	6	6	6
3	0	2	2	2	10	12	12	12	12	12	12	12	12	12	12	12
4	0	2	3	3	10	**12**										
5	0															

$$m[i,j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ \max\{m[i-1, j - w_i] + v_i, m[i-1, j]\} & \text{if } w_i \le j \\ m[i-1, j] & \text{otherwise} \end{cases}$$

$$v[1..5] = \{4, 2, 10, 1, 2\}$$

$$w[1..5] = \{12, 1, 4, 1, 2\}$$

$$W = 15$$

	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4	4
2	0	2	2	2	2	2	2	2	2	2	2	2	4	6	6	6
3	0	2	2	2	10	12	12	12	12	12	12	12	12	12	12	12
4	0	2	3	3	10	12	13	13	13	13	13	13	13	13	13	13
5	0	2	3	4	10	12	13	14	15	15	15	15	15	15	15	**15**

Pseudocode

⟨*Recursive cases*⟩

Time per table entry = $\boldsymbol{O(1)}$
Total time = $\boldsymbol{O(nW)}$

for $i = 1, \ldots, n$:
    for $j = 0, \ldots, W$:
        if $j \geq w[i]$:
            $m[i, j] \leftarrow \max(m\big[i-1, j-w[i]\big] + v[i], m[i-1, j])$
        else:
            $m[i, j] \leftarrow m[i-1, j]$

Question 4

We have n cents and need to get change in terms of denominations $d_1, d_2, \ldots, d_k$. Goal is to use the fewest total number of coins.

Example: If denominations are 25c, 10c, and 1c, then solution for n =30c should be 10c+10c+10c.

Let $M[j]$ be the fewest number of coins needed to change j cents. Write a recursive formula for $M[j]$ in terms of $M[i]$ with $i < j$.

Question 5

Using the above, derive a DP algorithm to compute the minimum number of coins of denomination $d_1, \ldots, d_k$ needed to change n cents.

Dynamic Programming algorithm paradigm

- Expressing the solution recursively
- Overall there are only polynomial number of subproblems
- But there is a huge overlap among the subproblems. So the recursive algorithm takes exponential time (solving same subproblem multiple times)
- So we compute the recursive solution iteratively in a bottom-up fashion (like in case of Fibonacci numbers). This avoids wastage of computation and leads to an efficient implementation

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